

Hollow-Structured $\text{Ti}_3\text{C}_2\text{T}_x$ Composite Fibers for Enhanced Pressure–Humidity Bifunctional Sensing in Febrile Convulsion Detection

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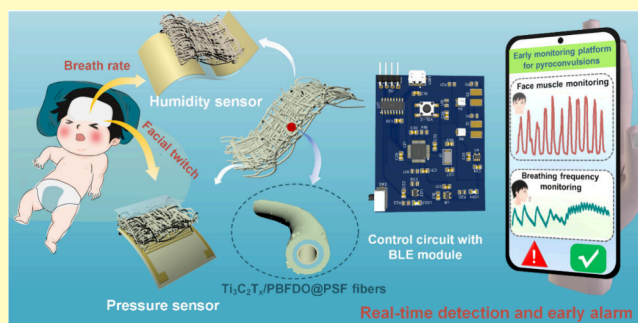
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ABSTRACT: Wearable bifunctional sensors demonstrate considerable potential for implementation in early warning systems targeting febrile convulsions. However, significant challenges persist in simultaneously achieving real-time and precise detection of dual sensing modalities with accurate signal discrimination for clinical diagnosis. Here, we have fabricated hollow-microstructure $\text{Ti}_3\text{C}_2\text{T}_x$ /poly(benzodifurandione) composite fibers for synergistic optimization of humidity–pressure sensing performance through rational microstructure engineering of the sensitive layer coupled with the dynamic reconfiguration of conductive pathways. The bifunctional sensor provides high sensitivity (168.72 kPa^{-1}), a low detection limit (0.67 Pa) for pressure sensing, rapid response/recovery time ($3 \text{ s}/20 \text{ s}$), and low humidity hysteresis ($2.18\% \text{ RH}$) for humidity sensing. Integrated with a control circuit, a febrile convulsion early alarm system is established for real-time monitoring of muscle activities and breathing rate by sending an early alarm to the mobile terminal when in abnormal conditions. This work develops a new avenue for prompt treatment of intelligent healthcare.

KEYWORDS: bifunctional sensor, hollow structure fibers, pressure and humidity sensing, $\text{Ti}_3\text{C}_2\text{T}_x$ MXene, early alarm system



Febrile convulsion, a common emergency case among humans from 6 months to 5 years old, is primarily caused by infections and has occurred in about 2% of children according to the survey.^{1–3} The typical symptoms of febrile convulsions involve fever, unconsciousness, partial muscle twitching, and shortness of breath and can even lead to severe brain damage and seizures unless treated promptly.^{4,5} Therefore, it is extremely crucial to establish an early system for the diagnosis and alarm of febrile convulsions. They are now primarily determined by inspection, laboratory examinations, and medical imaging examinations, which fail to provide real-time, continuous detection of patient conditions.

With the exponential advancement of flexible electronics, flexible sensors provide a novel potential for the detection and early alarm of febrile convulsions as a result of miniaturization, portability, and continuous and dynamic detection.^{6–9} Monitoring physiological information on febrile convulsions such as breathing rate and muscle activities in real time has the potential to revolutionize health care for patients. As essential physiological information, muscle activities could be readily detected by pressure sensors.^{10,11} The tiny and rapid muscle activities presented a challenge on detection limit, response/recovery time, and sensitivity for pressure sensors.^{12–14} Likewise, the breathing rate can be monitored by humidity

sensors with rapid response/recovery time and low hysteresis.^{15,16} The integration of flexible sensors with control circuits with wireless transmission enables the construction of a disease diagnosis platform for real-time monitoring of physiological conditions through mobile terminals.^{17,18} However, the complex diagnostic attributes of febrile convulsions, such as shortness of breath, muscle twitching, and fever, necessitate the acquisition of multiple sensing signals. The establishment of a diagnostic system for febrile convulsions, depending on the detection and threshold analysis of pressure and humidity signals, can prevent the risk of misdiagnosis induced by dynamic variations of single physiological parameter monitoring.

Thus, there exists an imperious need for a low detection limit for slight variations in muscle activities as well as a low hysteresis to detect rapidly variable breathing rates. For

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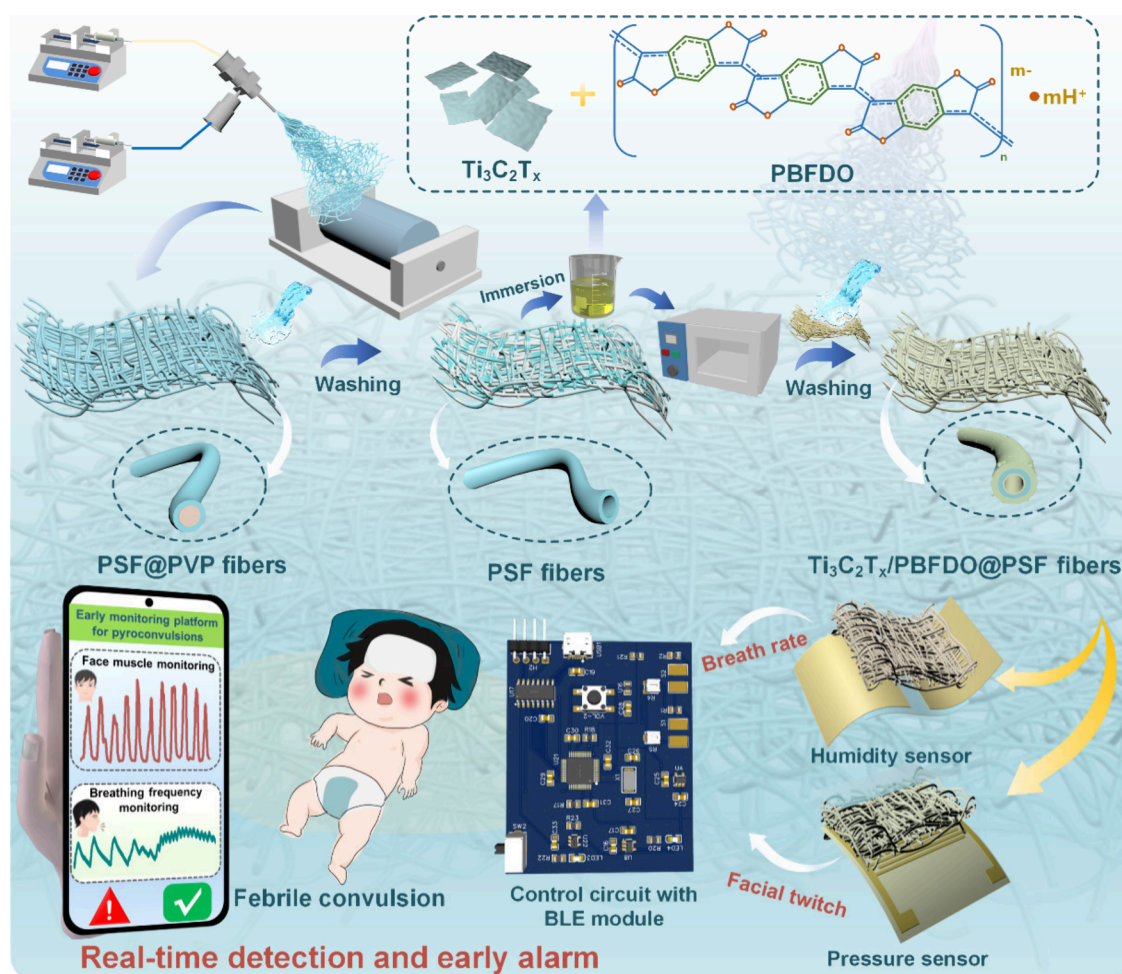


Figure 1. Schematic fabrication of the $\text{Ti}_3\text{C}_2\text{T}_x$ /PBFDO@PSF pressure–humidity bifunctional sensor for the early alarm of febrile convulsions.

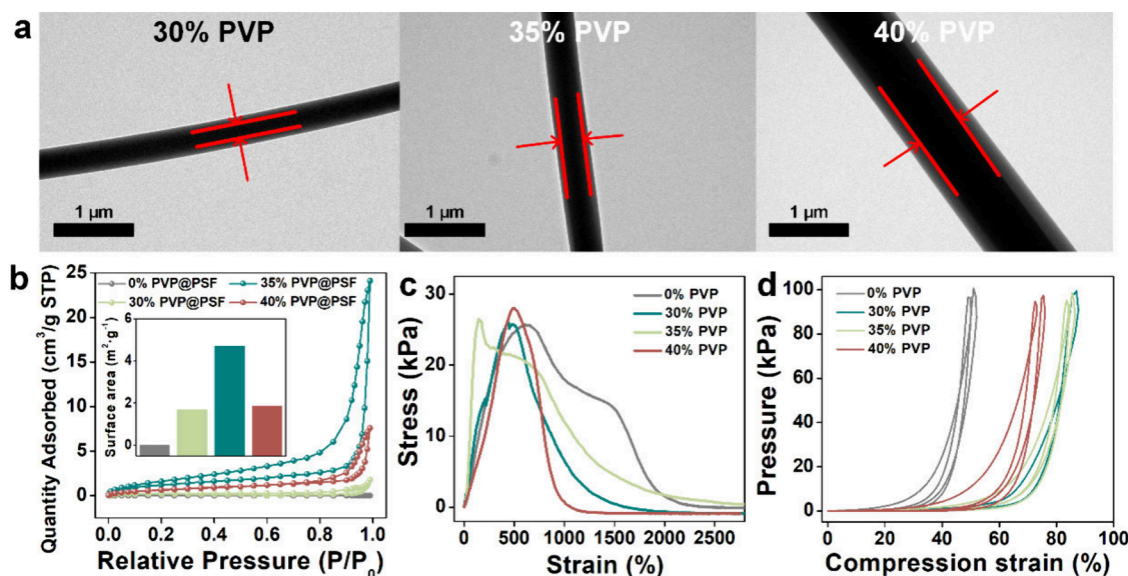


Figure 2. Characterization of $\text{Ti}_3\text{C}_2\text{T}_x$ /PBFDO@PSF with different PVP concentrations. (a) TEM images of PVP@PSF with different PVP concentrations. (b) N_2 adsorption–desorption isotherms of PSF@PVP with different PVP concentrations. Stress–strain curve of $\text{Ti}_3\text{C}_2\text{T}_x$ /PBFDO@PSF with different $\text{Ti}_3\text{C}_2\text{T}_x$ /PBFDO concentrations: (c) tensile/release and (d) compression/release.

pressure sensing, enhancing sensitivity is primarily implemented through the construction of microstructure-sensitive layers, which facilitates the formation of hierarchical

conductive network structures, regulating the generation of conductive pathways and improving the electronic transport capabilities of the sensor.^{19,20} According to the Persson contact

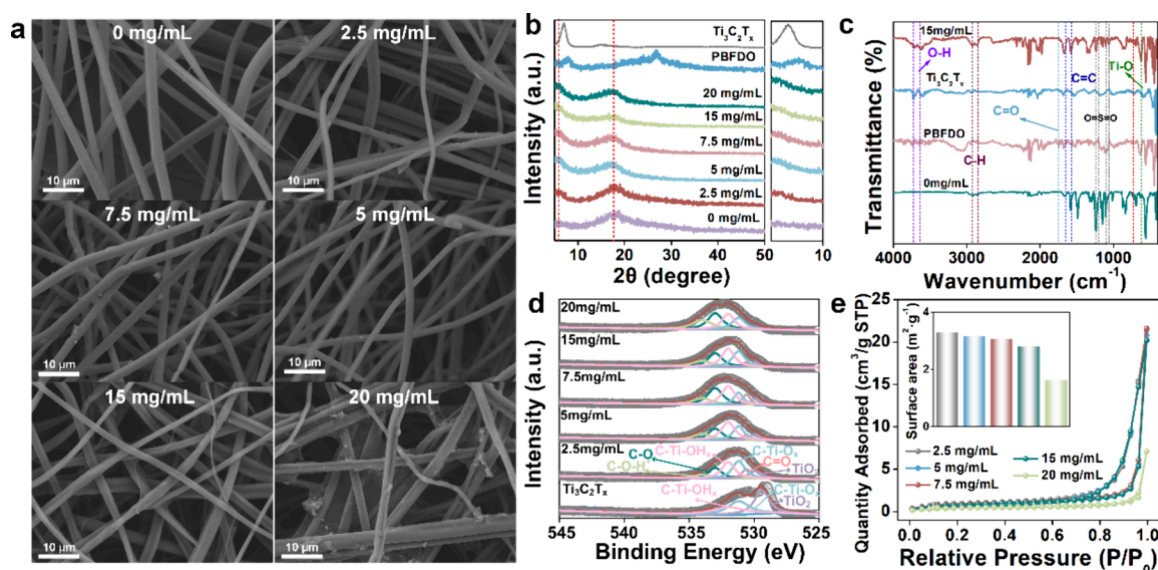


Figure 3. Characterization of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$. (a) SEM images of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with different $\text{Ti}_3\text{C}_2\text{T}_x$ concentrations. (b) XRD patterns of $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheet, PBFDO, and $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with different concentrations. (c) FTIR spectra of $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheet, PBFDO, and 15 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$. (d) O_{1s} XPS spectra of $\text{Ti}_3\text{C}_2\text{T}_x$ and $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with different concentrations. (e) N_2 adsorption–desorption isotherms of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with different concentrations.

theory, the detection limit of a pressure sensor is related to the compressive modulus of the sensitive layer.²¹ The sensor can achieve an ultralow detection limit by regulating the microstructure of the sensitive layer with reduced compressive modulus. For humidity sensing, the formation of heterojunctions within the sensitive layer creates electron depletion layers, which enhances H_2O adsorption, thereby enabling high sensitivity.^{22,23} Hollow-microstructure fibers are promising for synergistic optimization of humidity–pressure sensing performance by virtue of the superior surface area, lower compression modulus, and more active sites. $\text{Ti}_3\text{C}_2\text{T}_x$ MXene, a p-type 2D material with excellent mechanical strength, and extraordinary conductivity²⁴ provides superior performance in gas sensing,^{25,26} pressure sensing,²⁷ and humidity sensing.²⁸ As an n-type conductive polymer, poly(benzodifurandione) (PBFDO) can be involved in heterojunction construction with $\text{Ti}_3\text{C}_2\text{T}_x$ for humidity sensing and multilevel conductive pathways for pressure sensing.

Herein, a wearable pressure–humidity bifunctional sensor based on a $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with a hollow structure was fabricated. The hollow structure fibrous sensing layer brings more contact area, a lower compression modulus for pressure sensing, and more active sites for humidity sensing, leading to higher sensitivity and lower detection of pressure and better humidity response. It can precisely catch tiny and rapid muscle activities and record rapid breathing rates. Besides, an early detection and alarm system for febrile convulsion based on the bifunctional sensor and control circuit with a BLE module can display the detected muscle twitch and breathing rate curves on mobile terminals in real time. This system provides timely alarms on mobile terminals when physiological conditions exceed normal limits. This work holds great promise for the early alarm of febrile convulsions and intelligent healthcare.

RESULTS AND DISCUSSION

Preparation and Characterization of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ The fabrication process of the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ composite membrane was demonstrated in Figure 1. The few-

layered $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheet belonging to a hexagonal crystal system was synthesized (Figure S1). The PVP@PSF fibers were fabricated by electrospinning that exhibited a typical core–shell structure (Figure 2a). With the concentration of PVP increasing, the diameter of fibers increased. Compared with solid structure fibers, the fibers with hollow structures demonstrated a more specific surface area (Figure 2b). With the PVP concentration increasing from 0 to 35 wt %, the surface area increased from 0.0064 to $4.7 \text{ m}^2 \cdot \text{g}^{-1}$. The mechanical properties of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ were demonstrated by compression–relaxation and strain tests. As shown in Figure 2c and 2d, with PVP concentration increasing from 0 to 35 wt %, the Young's modulus and max compression strain increased. $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with 35 wt % PVP exhibited the highest Young's modulus of 0.235 kPa, and it also showed the largest compression strain under the same load pressure.

SEM images of the fibrous membrane (Figure 3a) revealed the morphology evolution of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with increased $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ concentrations (2.5 mg/mL, 5 mg/mL, 7.5 mg/mL, 15 mg/mL, and 20 mg/mL). $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with 2.5, 5, 7.5, 10, and 15 mg/mL showed a smooth surface and $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ was distributed on the surface of the fiber and between the fibers according to the elemental mapping images (Figure S2). With $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ concentration increasing to 20 mg/mL, the surface of the fiber got rougher and more $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ was randomly distributed between the fibers.

The structure of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ was characterized by XRD, XPS, and FTIR. The XRD pattern demonstrated the shift of the (002) peak of $\text{Ti}_3\text{C}_2\text{T}_x$ toward the low-angle direction and a significant decrease in intensity, which illustrated the layer spacing of the $\text{Ti}_3\text{C}_2\text{T}_x$ layer was expanded due to the introduction of PBFDO (Figure 3b). Compared with the XRD pattern of PBFDO, the crystal phase of PBFDO in $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ disappeared. It was possibly related to the formation of hydrogen bonding between $\text{Ti}_3\text{C}_2\text{T}_x$ and PBFDO that interfered with the regular arrangement of

conductive chain segments and reduced crystallinity.²⁹ The formation of hydrogen bonding between $\text{Ti}_3\text{C}_2\text{T}_x$ and PBFDO was further confirmed by the broadening of FTIR peaks around 3612 cm^{-1} ³⁰ (Figure 3c). Moreover, the vibration peaks of $\text{Ti}-\text{O}$ at 602 cm^{-1} , $\text{O}=\text{S}=\text{O}$ at 1245 cm^{-1} , 1106 cm^{-1} , and 738 cm^{-1} , and $\text{C}=\text{O}$ at 1740 cm^{-1} could be observed in the FTIR spectra, indicating the successful preparation of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ ^{31,32} (Figure S3). The O_{1s} peak of XPS revealed that the area of the TiO_2 phase of $\text{Ti}_3\text{C}_2\text{T}_x$ was reduced with the involvement of PBFDO, which indicated the oxidation process was prevented with the formation of hydrogen bonds³³ (Figure 3d and Table 1).

Table 1. $[\text{TiO}_2]/[\text{O}]$ Peak Area Ratio of $\text{Ti}_3\text{C}_2\text{T}_x$ and $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with Different $\text{Ti}_3\text{C}_2\text{T}_x$ Concentrations

Sample	$[\text{TiO}_2]/[\text{O}]$
$\text{Ti}_3\text{C}_2\text{T}_x$	0.51
2.5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$	0.13
5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$	0.09
7.5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$	0.07
15 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$	0.05
20 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$	0.03

Nitrogen adsorption–desorption was employed to investigate the pore structure of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ as shown in Figure 3c. The nitrogen adsorption–desorption isotherms of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with different concentrations were type II isotherms that were multilayer reversible adsorption on nonporous or macroporous surfaces.^{34,35} In the comparison of PSF@PVP with 35 wt % PVP, the surface area of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ was slightly decreased (Figure 3e). With the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ concentration increased to 35 wt %, the

surface area decreased slightly, and when the PVP concentration reached 40 wt %, the surface area notably declined.

Pressure Sensing Performance. The $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor ($8\text{ mm} \times 10\text{ mm}$) was fabricated so that the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ membrane was fixed on the PI/Au interdigital electrode and was simultaneously encapsulated on the top with a waterproof polyurethane (PU) (Figure 4a and Figure S4). The size of the pressure sensor was $8 \times 10\text{ mm}$ (Figure S5a). The strain behavior of the pressure sensor under different load pressures was demonstrated in Figure 4b. The resistance of the pressure sensor can be equated to the series connection of bulk resistance of the sensitive layer (R_F), the variable resistance (R_{C1}) of the sensitive layer, the bulk resistance of the electrode layer (R_E), and the variable contact resistance between it and the electrode layer (R_{C2}). Moreover, the contribution of R_E to the total sensor resistance was negligible due to the orders of magnitude difference with R_F , R_{C1} , and R_{C2} . In the initial (unloaded) state, less conductive pathways were established between the loosely distributed fibers, and the conductive fibers had only slight contact with the electrode layer, where only a small number of electrons flowed through the interface. With the load pressure increased, the contact area of conductive fibers and that between the sensitive layer and electrode enhanced, where R_{C1} and R_{C2} decreased, resulting in more electrons. When the load pressure reached the maximum, the sensitive layer was fully in contact with the electrode layer and the conductive fibers were fully compacted, where R_{C1} and R_{C2} were lowest and most electrons flowed through the interface.

During this process, $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ fibers with a hollow structure exhibited enhanced sensitivity, owing to the larger specific surface area and superior deformability. As the PVP concentration increased from 0 to 35 wt %, the sensitivity of the pressure sensors steadily improved, reaching a maximum

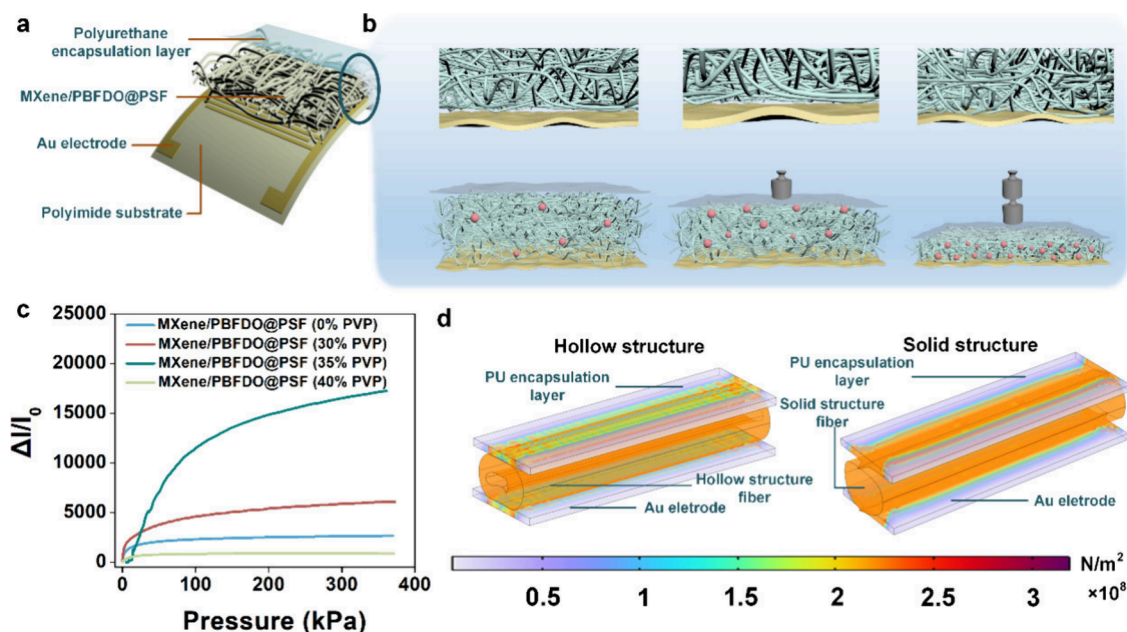


Figure 4. Piezoresistive effect of the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor. (a) Schematic of the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor. (b) Illustration of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor deformation states and the electron transport process under different pressures. (c) Response–pressure curves of the 15 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor with different PVP concentrations. (d) The compressing deformation mechanism of the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor with a hollow structure and solid structure is explained by FE simulation. The color contours represent the microscopic strain of the aerogel structure during the deformation process.

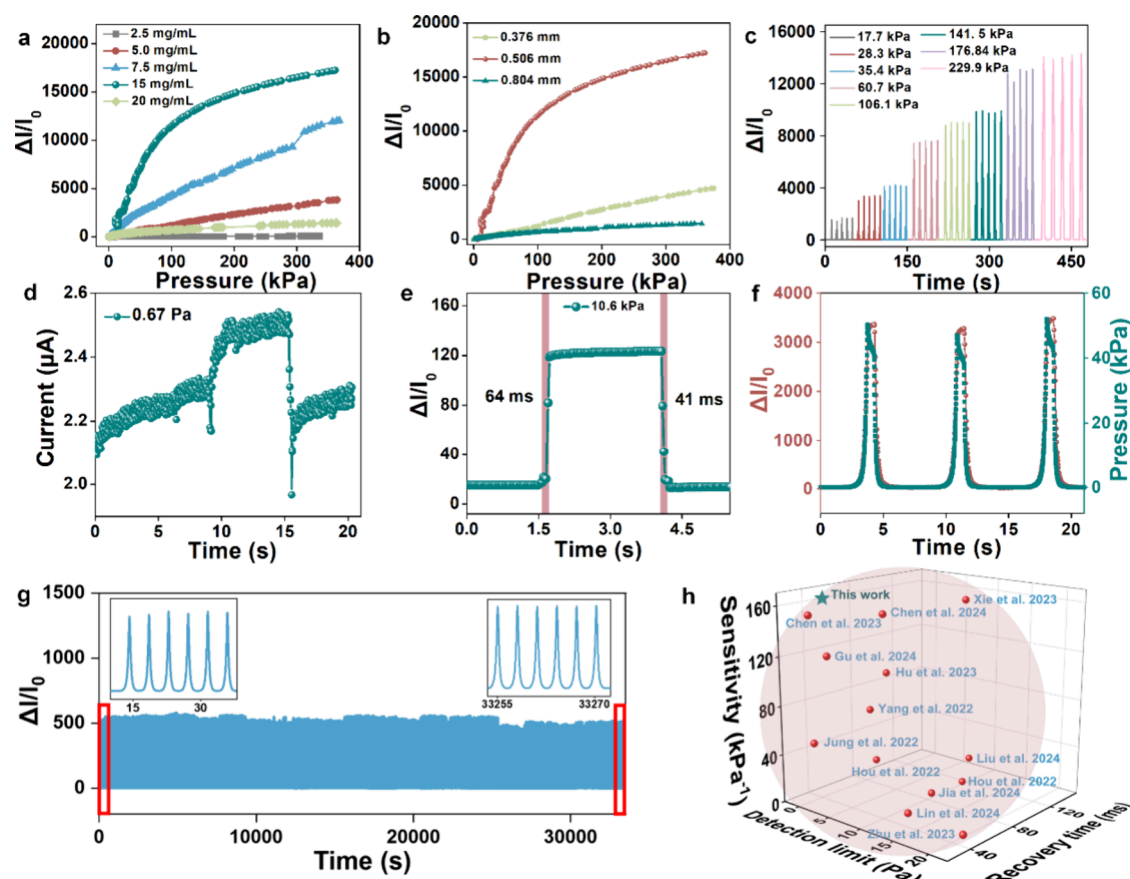


Figure 5. Pressure sensing performances of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$. (a) Response–pressure curves under various load pressures of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with different concentrations. (b) Response–pressure curves under various load pressures of 15 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with different thicknesses. (c) Relative current changes of 15 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ under load pressures of 17.7 Pa to 229.9 kPa. (d) The instantaneous response of I – t and P – t curves. (e) Response and recovery time at a pressure of 10.6 kPa. (f) The detection limit of the 15 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor. (g) Durability test under 10000 loading/unloading cycles at a pressure of 15.91 kPa; inset shows the enlarged curve for the first 6 cycles and last 6 cycles. (h) Performance comparison of piezoresistive sensors.

at 35 wt % (Figure 4c). This enhancement can be attributed to the largest specific surface area and strongest deformability compared with others, which contributed to greater changes in the contact area and consequently generated more conductive pathways under the same pressure. Moreover, when the PVP concentration was further increased to 40 wt %, both the specific surface area and mechanical properties deteriorated, leading to a sharp decrease in the formation of conductive pathways under pressure, and thus the sensitivity dropped obviously. The finite element (FE) simulation result further confirmed that the contact area change of the hollow structure was larger than that of the solid structure under equal stress (Figure 4d and Figure S6). Thus, $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ with 35 wt % PVP was preferred for the next measurement and application.

The pressure sensing performances of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ were measured at various levels of load pressure. The 15 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ exhibited the highest sensitivity (Figure 5a). The sensing mechanism was primarily related to the number of conductive pathways between the sensitive layer and electrode and inside the sensitive layer before and after pressure loading and unloading, where the conductive pathways were associated with the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ concentration and surface area. For 2.5, 5, and 7.5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$, due to the lower $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ concentration, the number of conductive pathways

formed after external pressure loading was not enough (Figure S7a). With the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ concentration increased from 2.5 to 15 mg/mL, the sensitivity was increased (Figure S7a–S7d). When the concentration of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ reached 20 mg/mL, the sensitivity started to decrease that the conductive pathways were enough although under no external pressure due to the excessive $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ and low surface area (Figure S7e). And the thickness of the sensitive layer also has a major effect on the sensitivity of the sensor. The pressure sensor with a 0.506 mm thickness sensitive layer exhibited the highest sensitivity (Figure 5b and Figure S7f). The 15 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor was selected for subsequent tests and applications. As shown in Figure S7d, the sensitivity was 164.72 kPa^{-1} in the range of 0–60 kPa and 59.36 kPa^{-1} in the range of 60–141 kPa with 0.980 and 0.977 linearity, respectively. However, when the external pressure was greater than 141 kPa, the response still grew as the pressure increased, but the sensitivity decreased to 17.62 kPa^{-1} .

As the external pressure grew from 17.7 to 229.9 kPa, the relative current change grew, while the response to the same pressure was stable (Figure 5c). And it can also detect extremely tiny pressure fluctuations down to 0.67 Pa (Figure 5d). The response/recovery time was 64 and 41 ms under 10.6 kPa (Figure 5e). The low hysteresis of 12.36% is shown in Figure S8a, which was further demonstrated by the consistency

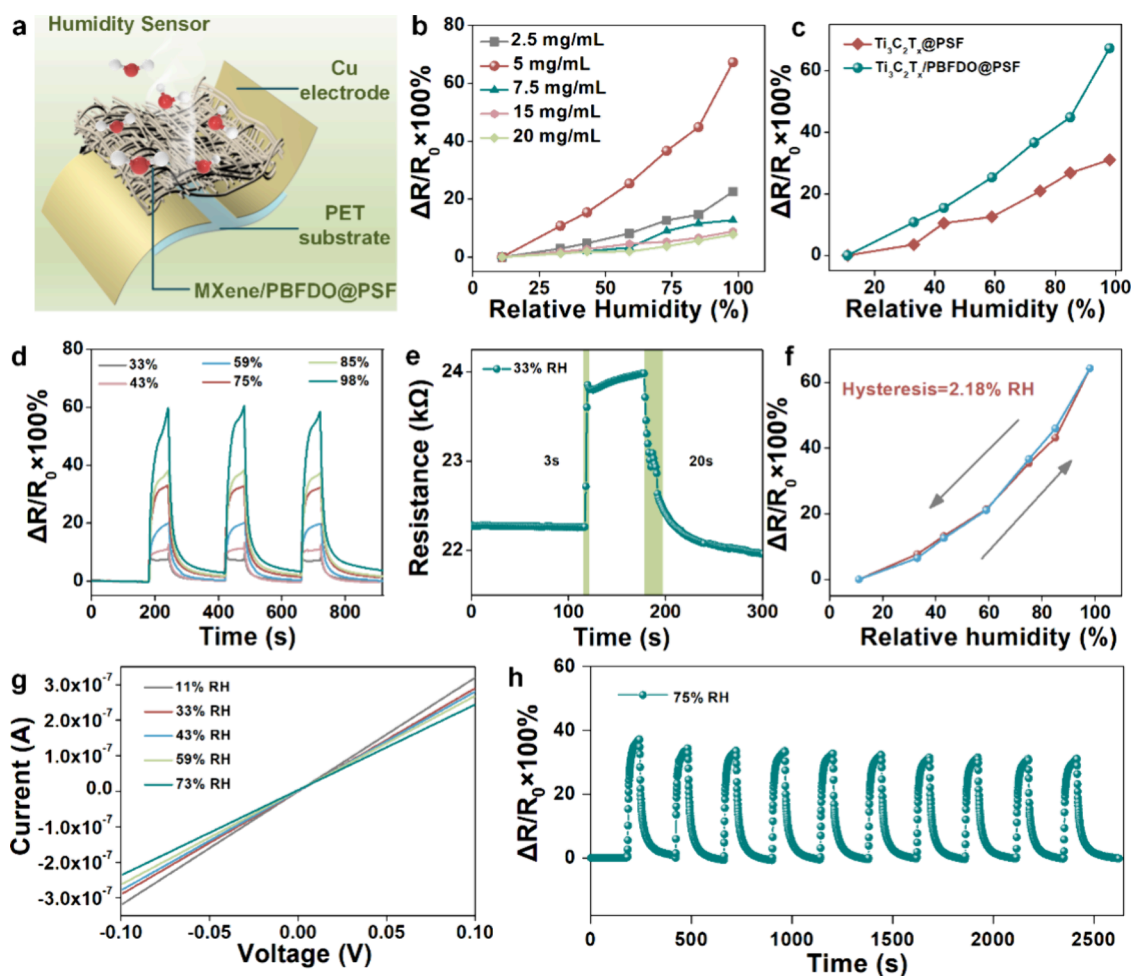


Figure 6. Humidity sensing performances of the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ sensor. (a) Schematic of the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ humidity sensor. (b) The sensitivity for humidity sensing of the 5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ humidity sensor. (c) The sensitivity for humidity sensing of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ and $\text{Ti}_3\text{C}_2\text{T}_x@\text{PSF}$ humidity sensors. (d) Dynamic response–recovery curve of the 5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ humidity sensor to different relative humidity. (e) The response and recovery time of the 5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ humidity sensor under 59% RH. (f) I – V curves of the 5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ humidity sensor under different relative humidity. (g) The humidity hysteresis of the 5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ humidity sensor. (h) The repeated response–recovery curve of the 5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ humidity sensor between 11% RH and 75% RH at 25 °C. (i) The real-time monitoring curve of respiratory rate.

between the output response signal and input external pressure when the external pressure was loaded and unloaded (Figure 5f). The excellent resolution of pressure sensing was confirmed by the obvious electrical signal (Figure S8b). The durability of the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor was tested by repeated pressure loading and unloading cycles under 15.91 kPa, exceeding 10000 cycles (Figure 5g). By comparing the first and last six cycles, the current change showed no attenuation. As demonstrated in Figure S9a, the initial current under no load and the response at a pressure of 15.91 kPa were stable over 31 days, indicating excellent long-term stability. Furthermore, for the illustration of the humidity stability, the pressure sensor encapsulated by waterproof PU was exposed for 10 min to environments with 11% RH, 33% RH, 43% RH, 59% RH, 75% RH, and 85% RH. The initial resistance of the pressure sensor was stable as well (Figure S9b). These results demonstrated the remarkable long-term stability and environmental robustness of the pressure sensor, which indicated the strong potential for reliable operation in different practical applications. In summary, this pressure sensor demonstrated high sensitivity, low detection limit, and rapid recovery and

also outperformed the previously reported piezoresistive sensors as shown in Figure 5h.^{36–48}

Humidity Sensing Performance. The humidity sensors (22×15 mm) were fabricated by fixing the fibrous sensitive layer on a polyethylene terephthalate (PET) substrate with Cu electrodes (10×15 mm) (Figure 6a and Figure S5b). The dynamic relative resistance change of the humidity sensor grew with relative humidity increasing from 33% to 98% RH (Figure 6b). Among those, 5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ exhibited the highest humidity sensitivity. Compared with the $\text{Ti}_3\text{C}_2\text{T}_x@\text{PSF}$ humidity sensor, it demonstrated a higher humidity response owing to the introduction of n-type conducting polymer PBFDO (Figure 6c). Figure 6d demonstrates the response and recovery curve of the humidity sensor under various relative humidity. With the relative humidity growing from 11% to 98% RH, the humidity sensor generated a distinguished response/recovery signal. It also exhibited rapid response/recovery times of 3 s and 20 s to 33% RH (Figure 6e). The humidity hysteresis was as low as 2.18% RH (Figure 6f and Figure S10). The excellent linear relation of the current–voltage (I – V) curves from -0.1 to 0.1 V was measured under different relative humidity, indicating the

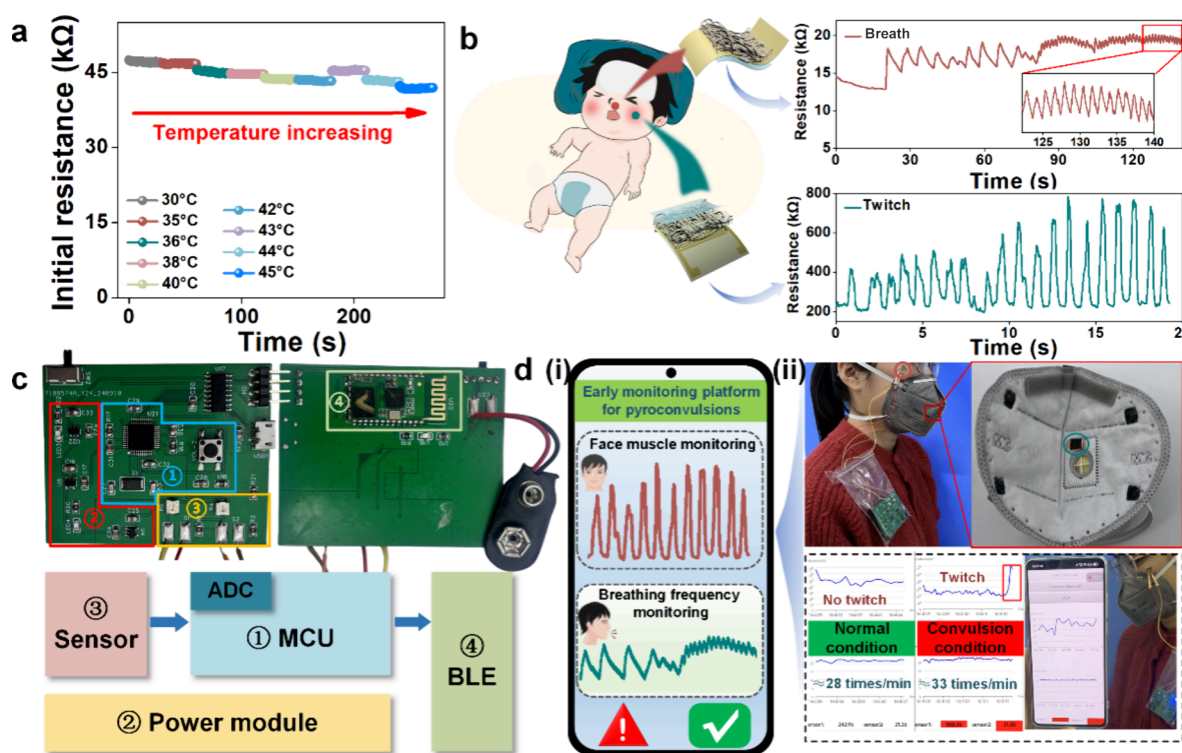


Figure 7. Wearable febrile convulsion alarm system for monitoring of breathing frequency and facial twitch. (a) The initial resistance of the sensitive material within the range of normal and fever temperatures of the human body. (b) The relative resistance changes of the humidity sensor and the pressure sensor when breath rate increases and facial twitching occurs. (c) Schematic illustration of the wearable febrile convulsion alarm system. (d) Flexible pressure–humidity bifunctional sensors integrated into the control circuit for the detection of breathing rate and muscle movements.

excellent ohmic contacts between the 5 mg/mL $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ and Cu electrodes (Figure 6g). The solid durability of the humidity sensor was demonstrated by 10 cycles of relative humidity increasing and decreasing from 11% RH to 75% RH (Figure 6f).

Early Alarm System of Febrile Convulsions. Febrile convulsions, a common pediatric emergency for children from 6 months to 5 years old, may lead to epilepsy and even severe brain damage if not detected and treated promptly. Therefore, it is extremely critical to provide real-time, continuous detection and early alarm of febrile convulsions to assist physicians in providing timely diagnoses and delivering appropriate emergency treatment. The common symptoms of febrile convulsions include fever, loss of consciousness, localized muscle twitching, foaming in the mouth, and rapid breathing. Pressure–humidity bifunctional sensors with an ultralow detection limit of pressure sensing and rapid response/recovery time of humidity sensing that can monitor tiny muscle twitching and breathing rate are essential for early alarm of febrile convulsions.

The application of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ bifunctional sensors for the early alarm of febrile convulsions was further demonstrated. The initial resistance of sensitive materials was examined at different simulated human body temperatures from 30 to 45 °C. The electrical properties of the sensitive materials have not been dependent on temperature variations. Moreover, the response values of humidity and pressure sensors are essentially independent within the range of human body temperature variation from 36 to 40 °C. As demonstrated in Figure S11a, the humidity response was stable, and the baseline decreased slightly, which was attributed to the

enhanced desorption of H_2O molecules caused by the temperature increase. The response of the pressure sensor was stable as the temperature varied as well (Figure S11b). Thus, the temperature of the human body under normal and feverish conditions did not generate any error in the sensor (Figure 7a).

When the bifunctional sensors were attached to the facial muscles and near the mouth and nose, tiny muscle twitching and quick breathing frequencies could be detected (Figure 7b). Moreover, the resistance change curve of nasal and oral breathing could be accurately detected, and both baseline and peak were relatively stable, which indicated that the breathing rate could be precisely recorded no matter which breathing mode was adopted (Figure S12). Integrated $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ -based pressure/humidity bifunctional sensors with a printed circuit board (PCB) with a size of $5 \times 4 \text{ cm}^2$, an early alarm system for febrile convulsion with real-time monitoring and alerting on mobile terminals, were developed (Figure 7c).

The real-time response curves of pressure and humidity related to the muscle twitch and breathing rate, respectively, could be visualized through the application page on a mobile phone, and the control program could determine whether pressure variations were generated and breathing rate ranges, as the wireless Bluetooth module was connected to the mobile terminal as shown in Figure 7d (i). Figure 7d (ii) demonstrates the actual photograph of the detection process. The pressure sensor was adhered to the facial muscles, and the humidity sensor was attached to the facemask to detect muscle motions and breathing rate. The normal breathing rate of a healthy toddler is 25–30 times/min. When breathing rate is normal

and no muscle twitching occurs, the breathing humidity curve and muscle activity curve will be displayed in real time on the mobile interface without sending any alarms (Supplementary Movie 1). When the breathing rate is lower than 25 times/min or higher than 30 times/min or when partial muscle twitching occurs, the mobile phone application linked to the Bluetooth module can send an alarm in real time (Figure 7d, Figure S13, and Supplementary Movies 2, 3, and 4).

CONCLUSION

In brief, through ingenious structural design and the devising of heterojunctions for sensitive materials, a $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ bifunctional sensor for pressure and humidity sensing was fabricated to precisely monitor weak muscle activities and rapid breathing rate. The sensing performances of pressure and humidity sensing were optimized for precise monitoring of physiological signals with 164.72 kPa^{-1} sensitivity, 0.67 Pa detection limit for pressure sensing, 3 s and 20 s response/recovery times, and ultralow hysteresis ($2.18\% \text{ RH}$) for humidity sensing. Therefore, the bifunctional sensors with high performance were integrated into the wearable early alarm system for febrile convulsion so that the muscle activities and breathing rate can be displayed on the mobile terminal and give an alarm for abnormal conditions. This high-performance bifunctional sensor provides a bright potential in intelligent healthcare in the future.

EXPERIMENTAL SECTION

Preparation of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene Nanosheet. $\text{Ti}_3\text{C}_2\text{T}_x$ MXene was synthesized by wet etching using a LiF/HCl solution to remove the Al layer from Ti_3AlC_2 (Jilin 11 Technology Co., Ltd., China). Initially, 1 g of LiF (Aladdin, $>99.99\%$) was dissolved and stirred vigorously for 15 min in 40 mL of 12 M HCl (Xilong Scientific Co., Ltd.) in a Teflon-lined vessel. Then, 1 g of Ti_3AlC_2 was slowly added into the LiF/HCl mixed solution, which reacted at 45°C in a water bath for 24 h . After etching, the resultant mixture underwent repeated centrifugation for 8 min at a speed of 8000 rpm , continuing until the pH of the supernatant reached $6-7$. Subsequently, the clay mixture was dispersed ultrasonically in an ice bath at 350 W for 1 h . Ultimately, $\text{Ti}_3\text{C}_2\text{T}_x$ MXene nanosheets were obtained through centrifugation at 10000 rpm for 10 min .

Preparation of Hollow Structure PSF Fibrous Membrane. PVP@PSF fibrous membranes with different PVP concentrations were prepared by electrospinning. The electrospinning precursor solution contained $27 \text{ wt } \%$ PSF (Aladdin, $M_w = 35000$) in N,N -dimethylacetamide (DMAc, Sinopharm Group Chemical Reagent Co., Ltd.) and tetrahydrofuran (THF, Sinopharm Group Chemical Reagent Co., Ltd.) ($1/1$, w/w) and a certain concentration (30 , 35 , and $40 \text{ wt } \%$) of polyvinylpyrrolidone (PVP, Aladdin, $M_w = 58000$) in ethyl alcohol (Xilong Scientific Co., Ltd.). The two precursor solutions were placed in 5 mL syringes with a coaxial nozzle and controlled by programmed syringe pumps with 1 mL/h (PSF) and 0.2 mL/h (PVP) feed rates, respectively, where PSF was selected as the shell material and PVP was employed as the core material. The high-voltage power supply provided 18 kV with a nozzle-to-collector distance of 15 cm . In the electrospinning process, the environment humidity and temperature were fixed at $40 \pm 2\%$, 20°C . Then the hollow structure PSF fibrous membrane was obtained through repeated washing with deionized water to remove the PVP layer.

Preparation of $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$. Different masses of the obtained $\text{Ti}_3\text{C}_2\text{T}_x$ MXene nanosheets powder (0.25 g , 0.5 g , 0.75 g , 1.5 g , and 2 g) were dissolved in deionized water. And different volumes of PBFDO solution (containing 0.25 , 0.5 , 0.75 , 1.5 , and 2 g of PBFDO) were added into the above solution, which was dispersed ultrasonically at 300 W for 10 min . Then the hollow structure PSF fibrous membrane was immersed in $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO}$ solution and

was left at room temperature in a vacuum oven for 2 h . The $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ was acquired by repeated washing with deionized water until the membrane did not discolor.

Characterizations. The morphology of the $\text{Ti}_3\text{C}_2\text{T}_x$ MXene nanosheet, $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$, and $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ pressure sensor was observed by scanning electron microscopes (FESEM, JEOL JSM-7900F) with an acceleration voltage of 5 kV . The morphology of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene and PVP@PSF fibers was observed by transmission electron microscopy (TEM; JEM 2100 F) with an acceleration voltage of 200 kV . The chemical components and structure over a range of $4000-400 \text{ cm}^{-1}$ were measured by a Fourier transform infrared spectrometer (FTIR, Nicolet iS10). The crystalline structures were characterized by wide-angle X-ray diffraction (XRD, Rigaku D/Max 2550) in the 2θ range from 5° to 50° with $\text{Cu K}\alpha$ radiation ($\lambda = 1.5418 \text{ \AA}$). The scanning rate was $5^\circ/\text{min}$. The chemical composition of the $\text{Ti}_3\text{C}_2\text{T}_x$ -based aerogel surfaces was examined by X-ray photoelectron spectroscopy (XPS, ESCALAB 250) with an X-ray source ($\text{Al K}\alpha h\nu = 1486.6 \text{ eV}$). The specific surface area and pore size distribution were measured by nitrogen adsorption/desorption isotherms through a Brunauer–Emmett–Teller device (BET, Autosorb-iQ-C).

Pressure-Sensing Performances. The different pressure conditions were applied by a compression testing machine with TRAPEZIUM X pressure-acquired software (AGS-X 500 N, Shimadzu, Japan). The real-time current of the pressure sensor was measured by connecting the digital multimeter (Keithley DMM7510), and the 0.1 V DC voltage was provided by a DC power supply (DPS-305BM). The response (R_i) was defined as the relative current change ($\Delta I/I_0$), where ΔI is the change of the current between the loading pressure state and unloading pressure state. The sensitivity (S) of the pressure sensor was defined as $\delta R_i/\Delta p$, where p refers to the intensity of the pressure. The electrical hysteresis was defined as the ratio of the area beneath the $\Delta I/I_0$ curves under loading and unloading.

$$\text{Hysteresis (\%)} = \frac{S_{\text{unloading}} - S_{\text{loading}}}{S_{\text{loading}}} \times 100\% \quad (1)$$

where S_{loading} and $S_{\text{unloading}}$ represent the integral area of the $\Delta I/I_0$ and pressure curves under pressure loading and unloading, respectively.

Humidity-Sensing Performances. The different humidity conditions were provided by LiCl , MgCl_2 , K_2CO_3 , NaBr , NaCl , KCl , and K_2SO_4 saturated solution flasks with the relative humidity of $11\% \text{ RH}$, $33\% \text{ RH}$, $43\% \text{ RH}$, $59\% \text{ RH}$, $75\% \text{ RH}$, $85\% \text{ RH}$, and $98\% \text{ RH}$, respectively. The LiCl saturated solution flasks with $11\% \text{ RH}$ humidity were employed as a dry air bottle. The real-time resistance of the $\text{Ti}_3\text{C}_2\text{T}_x/\text{PBFDO@PSF}$ humidity sensor was recorded by the digital multimeter (Fluke 8846A). The humidity response was defined as the relative humidity change rate ($\Delta R/R_0 \times 100\%$), where ΔR is the change of the resistance that was placed in a high relative humidity bottle for 1 min to $11\% \text{ relative humidity (RH)}$. During the measurement process, the humidity sensor was placed at $11\% \text{ RH}$ and different relative humidity levels for approximately 3 and 1 min . The response/recovery time was defined as the time needed to reach 90% of the total signal change from $11\% \text{ RH}$ to high (from high to $11\% \text{ RH}$) humidity.

FE Finite Element Simulation of the Stress Distribution. The COMSOL Multiphysics software was employed to conduct the finite element simulation. For simulating the stress distribution, the “Solid mechanics” module, coupled with the “Stationary” study, was set. First, the geometric model of the pressure sensor was built. The Au electrode layer and PU encapsulation layer were all set as a cuboid with a 200 mm width, 60 mm depth, and 5 mm height. The fiber with a solid structure was set to be a cylinder with a 20 mm radius, and that with a hollow structure was set as a different set of concentric cylinders with 20 mm and 8 mm radii. Additionally, three contact pair nodes were set between the bottom of the PU encapsulation layer and the top of the fiber, the top inner surface and the bottom inner surface of the fiber, and the bottom of the fiber and Au electrode layer, respectively. Then, the bottom boundary of the Au electrode layer was set as a fixed constraint node, and the top of the PU encapsulation

layer was set as prescribed displacement. The mesh was constructed on the models by the Free Quadrilateral feature node. Ultimately, the computation was executed to acquire the distribution of the stress and contact area when subjected to different pressures.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssensors.5c02521>.

Additional materials characterization including TEM image of $\text{Ti}_3\text{C}_2\text{T}_x$; SEM and FTIR spectra, EDS images of sensitive materials; the FE simulation result; sensitivity, hysteresis, and stability of the pressure sensor; the response–recovery curve and breathing curve of the humidity sensor; the photograph, temperature stability, and the operation demonstration of the system (PDF)

Operation process of febrile convulsion detection system with normal breathing rate and no muscle twitching (MP4)

Operation process of febrile convulsion detection system with normal breathing rate and muscle twitching (MP4)

Operation process of febrile convulsion detection system with shortness of breath and no muscle twitching (MP4)

Operation process of febrile convulsion detection system with shortness of breath and muscle twitching (MP4)

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Author Contributions

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Notes

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